RC2 Temporal Sample Protocol

Lab Supplies

Vial Preparation

- OCN
  - 40 mL amber vials w/ septa, marked at the ¾ point, no preservation
  - Labeled for triplicate sampling at 6 sites
  - i.e., from RC2_0001_ICR-1 to RC2_0001_ICR-3 (Site 1)…all the way through RC2_0006_ICR-1 through RC2_0006_ICR-3 (Site 6)

  FTICR (3 vials per kit)

  - 40 mL amber vials w/ septa, marked at the ½ point, pre-acidified w/ 10 µL 85% H_3PO_4
  - Labeled for triplicate sampling at 6 sites
  - i.e., from RC2_0001_ICR-1 to RC2_0001_ICR-3 (Site 1)…all the way through RC2_0006_ICR-1 through RC2_0006_ICR-3 (Site 6)
- IONS
  - 15 mL proteomic friendly white cap tubes, no preservation
  - Labeled for triplicate sampling at 6 sites
  - i.e., from RC2_0001_ICR-1 to RC2_0001_ICR-3 (Site 1)…all the way through RC2_0006_ICR-1 through RC2_0006_ICR-3 (Site 6)
- DIC
  - 40 mL amber vials, no marking, no preservation
  - Labeled for triplicate sampling at 6 sites
  - i.e., from RC2_0001_ICR-1 to RC2_0001_ICR-3 (Site 1)…all the way through RC2_0006_ICR-1 through RC2_0006_ICR-3 (Site 6)
- TSS
  - 6 - 2L brown Nalgene containers, rinsed with milli-Q water
  - Labeled for each site: RC2_0001_TSS-1 through RC2_0006_TSS-1
- DNA
  - Sterivex filter with luer lok
  - One filter for each site labeled RC2_0001_RNA-1 through RC2_0006_RNA-1

Six (6) sampling kits (pre-packaged by sampling date); each kit contains the following:

- 3 DIC vials
- 3 ICR vials
- 3 OCN vials
- 1 TSS bottle
- 1 Sterivex with luer lok
- Whirklpak with luer lok caps
- 2 Needles
- 2 mL epi tube with RNALater
- 1 - 3mL syringe
- 1 - 60 mL syringe

Extras Bag

- Gloves
- Filters luer and non luer
- Whirlpaks with luer caps
- Needles
- Syringes 3 mL and 60 mL
- Epis with RNA later
- Garbage bag
- “sharps” container

Extras Box

- Extra vials of all types used in the site kits (15 mL tubes, 40 mL vials, 2 L bottles)
- Extra filters
- Box of Ziploc bags
- Box of gloves
- Rope
- Tool bag
- Spare batteries
  - C (Manta, YSI, Ultrameter)
- Cooler and ice packs

Emergency Kit

- Satellite phone
- First Aid Kit
- KN95s
- Binder with map, ORMP, Field Permitting plan, parking pass
- Antibacterial hand cleaner
- Towels

Personal items to bring along

- Wading boots
- Safety vest
- Bug spray
- Lunch and snacks

Water Chemistry Sampling

- Water chemistry analytes (Triplicate sampling w/ the exception of TSS and microbial filter)

Core parameters	Storage Location
NPOC/TN	4 deg C
Ions	-20 deg C
FTICR	-20 deg C
DIC	4 deg C
TSS	4 deg C
RNA filter	-80 deg C

Sample Collection

- Filtered samples
  - Take the 60 mL syringe and pull a volume of water then discard to rinse. Repeat for a total of 3 times
  - Attach a needle to the labeled filter unit and then attach that to the 60 mL syringe. Push 5 mLs of water and discard that volume then continue to fill the vials in the following order:
    - DIC- Insert the needle into the vial and fill until it overflows. You need a total of 160 mL of water to go through. Fill slowly as to not create bubbles. At the end fill until there is a dome at the top of the vial. Flip upside down to ensure there are no trapped air bubbles.
    - Ions- Fill to the 10 mL line
    - FTICR- fill halfway to the pre-marked line
    - NPOC/TN- fill ¾ of the way to the pre-marked line
    - RNA Later Filter- Pump water through the filter until it has reached 1 L of water or until it has clogged. (A majority of the time it clogs before finishing filling all the vials). Then a syringe with air and use that to push the remaining water out of the filter. Cap one end with the male luer lok cap and use the 3 mL syringe+needle to pull up the 3 mL of RNAlater out of the epi tube. Remove the needle then push the RNAlater out of the 3 mL syringe and into the filter housing. Secure the other end with the female luer lok cap and invert the filter a couple times to ensure the RNA later has covered the entire filter. Place into a whirlpak and place on ice. Do not let water enter the whirlpak.
- Unfiltered sample
  - TSS sample collected- Dip the sample bottle into the river and rinse 3 times. For the actual sample, dip the bottle with the opening facing downstream as to not catch any large surface water debris. Fill completely.

*Place all samples on ice after collection

*If your filter clogs before you are done filling all the vials, grab a new filter and push 5 mLs of water again before filling the rest of the vials.